Empyrean Roots

Research, community operations and project decisions

Simon Mukooza | Co-founder and operations lead

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Turning a sustainability ambition into work people can carry out

Empyrean Roots is a regenerative enterprise co-founded by Kera and Simon Mukooza. Its research connects community priorities in Uganda with the practical requirements of sustainable materials, clean energy, measurement and enterprise development. This report presents my contribution: designing the inquiry, coordinating local facilitators, translating findings into operating requirements, and revising the plan as evidence changed.

The work extended beyond writing a business concept. It involved field questions, multilingual materials, remote coordination, local relationships, repeated planning decisions and expenditure over more than two years. Research and planning have continued since the initial 2023 work, with dated feasibility and design reviews in 2026.

Scope at a glance

- Six districts in the broader research program: Mbale, Busia, Mukono, Kayunga, Masaka and Jinja.
- Three-person core field team in the recorded 2023 community-delivery work.
- Five languages in the engagement framework: Luganda, Kiswahili, Runyankore, Runyarwanda and English.
- A deployed 12-question community questionnaire, supported by local proxies and several response-capture methods.
- More than $26,000 in R&D expenditure, reported by the founder, across more than two years. No independently audited total or category allocation is presented.
- Two-week R&D planning cycles, alongside dated reviews and decision gates.

What the reader can examine

The following pages explain geography and timing, field delivery, research methods, tools, spending categories, technical research and decisions. The closing section connects these lessons to Pro Project Manager (PPM), a separate project-management tool.

This is an account of research and delivered community-engagement work. Production facilities, carbon-credit issuance and measured environmental benefits are not presented as completed outcomes.

1. Geography, chronology and responsibility

Six named districts, with boundaries on the timeline

The broader research covered Mbale, Busia, Mukono, Kayunga, Masaka and Jinja. These are the locations I confirmed for this report. Earlier accounts recorded five districts; this six-place list is my current clarification of the broader program. District-by-district dates have not been reconstructed, so this report does not invent an itinerary or assign a particular language, participant total or expenditure to any district.

The geography of engagement must also be distinguished from site planning. Kayunga appears in later materials and production-planning research. A district's appearance in field research does not establish that a facility was secured, constructed or commissioned there.

Development over time

- March-May 2023: the recorded community-delivery pilot account, including remote coordination of three facilitators and multilingual materials.
- June 2023-December 2024: the foundational R&D and blueprint-development phase described in the project record.
- July 2026: dated reviews of the wider venture, a revised Kayunga first-stage approach, carbon-methodology screening and pre-layout requirements.
- October 2026: further conceptual campus-layout review and preparation of this public account.

The expenditure described here spans more than two years. It should not be assigned entirely to the short 2023 pilot. The dated reviews demonstrate continued investigation; they do not prove that every intervening fortnight has a preserved sprint record.

My operational responsibility

I wrote and selected the fieldwork questions, designed how responses would be captured, coordinated research through local proxies, prepared engagement materials and synthesized findings into the regenerative blueprint. I also worked on sequencing, operating requirements and the conditions that would need to be met before larger commitments.

The venture is a joint undertaking with Kera. This account identifies my operational contribution without claiming sole ownership of every strategic or design input. Local facilitators contributed language, context and judgment that could not be supplied remotely.

2. The community delivery system

A practical definition of a proxy

A proxy was a local person able to carry the work into a community, interpret context and return feedback. The role required more than forwarding a message. The liaison adapted the introduction, explained the material, listened for confusion and helped me understand what needed to change.

The delivery chain was: Simon prepares the inquiry and materials; the local liaison adapts and delivers; community participants respond; feedback returns for revision. Human judgment remained at the handoff between the centrally prepared material and the people using it.

A repeatable session kit

Each topic was supported by a facilitator script, Canva visual cards, a take-home sheet and a short feedback prompt: "What confused people?" Materials used short sentences, one idea per card, explanations of technical terms and scripts designed to be read aloud. Literacy and comfort with written forms were not assumed.

The framework covered Luganda, Kiswahili, Runyankore, Runyarwanda and English. Google Translate supported language work, with native-speaker review supplying the contextual check. Language coverage is a feature of the framework, not a claim that every item was delivered in all five languages in every district.

Coordination that matched the setting

WhatsApp groups and voice notes supported proxy coordination. WhatsApp video, Zoom and Google Meet provided routes for remote discussion. Paper forms remained an option when people were wary of phones; responses could then be typed into Sheets. The method adapted to participants' comfort instead of making a particular device a condition of participation.

The engagement procedures emphasized introductions through trusted leaders, listening before presenting, allowing local hosts to lead, context-appropriate conduct and leaving a contact route for questions. These were practices for the communities involved, not universal assumptions about Uganda.

Increasing local ownership

The intended progression was from my drafts, to liaison adaptation, to independent use of templates. I report that facilitators could run sessions independently within two months in the pilot work. This is a founder-reported delivery outcome, not an independently timed evaluation. The operational lesson was concrete: materials succeeded when people could use them without my constant presence.

3. How the research was collected and interpreted

Question design and participation

I designed and deployed a 12-question intake questionnaire to understand community priorities, including what people wanted addressed first and last. Participants included community leaders, boda boda drivers, working mothers, stay-at-home mothers and people from varied income groups. That breadth helped expose different concerns; it does not establish a representative sample.

The central research task was to turn responses into decisions about participation and sequencing. A technically attractive idea still needed to be understandable, locally relevant and feasible to introduce through the people available to deliver it.

Several routes for capturing responses

I used Google Forms, KoboToolbox/ODK and paper forms, with Excel and Google Sheets supporting structured work. XLSForm was relevant because it provides a spreadsheet-based approach to authoring forms for compatible collection tools. This explains the tool choice; it does not mean that a single integrated pipeline connected every platform. [1]

Paper responses were typed into Sheets when phone-based collection was unsuitable. The public report does not include identifiable responses or claim a verified transcription-error rate, complete digital audit trail or uniform collection method across all locations.

From a response to a revision

The recorded working loop was draft, localize, deliver, receive feedback and revise. A question that sounded clear in a remote draft could still be confusing when read aloud. The liaison's feedback therefore informed both the content and the process used to deliver it.

I synthesized the findings into the blueprint and used the research to reconsider the order of development. One recorded change was moving from an initial construction-blocks starting point toward addressing measurement, reporting and verification (MRV) earlier. Later feasibility work revisited the practical first stage again. A change in sequence was a research output, not evidence that either sequence had already been built.

Limits that keep the findings usable

No respondent count, response rate or statistical generalization is supplied here. Underlying participant data is not released with this case study. This report supports an account of questionnaire design, field coordination, qualitative learning and evidence-informed prioritization; it does not present a population survey or quantified impact evaluation.

4. Managing research as project work

Iterative planning with consequential decisions

I describe the R&D planning rhythm as two-week cycles: deciding what came first, what could wait and who owned the next piece of work. The historical account records six concurrent workstreams. Their original names and a complete cycle-by-cycle backlog are not reproduced here, so the research categories on the following page should not be read as the original sprint-board labels.

This is evidence of applied Agile planning in a research setting. It is not a claim of Scrum certification, a formally audited Scrum implementation or Agile management of an operating factory.

Tools used across the work

- Planning and coordination: Asana, Microsoft Project and Smartsheet; Excel and Google Sheets for supporting schedules, research and working records.
- Shared thinking and documents: Google Jamboard during the historical work, Google Docs, Microsoft Word and Microsoft 365.
- Field collection and communication: Google Forms, KoboToolbox/ODK, paper forms, WhatsApp and remote meeting tools.
- Materials and prototypes: ChatGPT and Claude.
- Data visualization: Looker.
- Sources and references: Zotero and Drive source folders, alongside standards and methodology documents.

These are tools I confirmed using. A list of tools alone does not establish a process: the stronger evidence is the research question, assigned work, returned finding and decision that followed.

What the retained decision record shows

Dated reviews preserved challenged assumptions, conditions for proceeding and alternatives if evidence did not support the preferred plan. The BARAZA reviews used structured, AI-assisted adversarial analysis. They were a way to question the proposal, not independent certification by an expert panel.

The preserved artifacts are strongest as evidence of decision gates and event-driven replanning. They do not constitute a complete archive of every two-week cycle. This distinction matters when explaining how planning cadence and actual decision timing interacted.

The management lesson is that a review must be allowed to change the plan. A schedule that survives every adverse finding unchanged is not demonstrating learning.

5. The breadth of the research

Community adoption and communication

This work examined who could introduce the initiative credibly, what people wanted prioritized, how materials should be explained and how local facilitators could gain independence. Its outputs included the questionnaire, delivery framework, scripts, visual cards and feedback loop.

Sustainable materials and operating requirements

The materials research considered construction-block production and the requirements of a potential brick center. Pre-layout work addressed product definitions, mixture assumptions, shift meaning, raw-material movement, curing, quality checks, dispatch and worker needs. These were requirements and questions for development, not an engineering-approved design or an operating production line.

Carbon measurement and methodology

Research examined MRV and candidate carbon-accounting routes, including the UNFCCC AMS-III.Z methodology, Verra materials and Gold Standard materials. The AMS-III.Z screening was an initial suitability investigation. Reading a methodology or registry does not establish project eligibility, registration or credit issuance. [2-4]

The questions included baseline assumptions, additionality, material composition, energy use, transport, testing and the cost and practicality of measurement. Research separated physical output from a crediting outcome: a quantity of blocks cannot simply be presented as an equivalent quantity of verified carbon credits.

Wider regenerative pathways

The blueprint also explored clean energy, biochar and organic materials. These broadened the operating concept and exposed dependencies between production, community participation and measurement. They remain research and development pathways in this account; no delivered output or environmental return is assigned to them.

Feasibility and site organization

The reviews tested the relationship between demand, capital commitments, timing and prospective carbon income. Later conceptual layout work examined movement, functional separation and site assumptions. Terrain, wind, site conditions and professional validation remained material questions. A conceptual layout helped surface requirements; it did not remove the need for surveys or qualified design work.

Together, these areas show the extent of the inquiry: community acceptance, technical requirements and commercial sequencing had to be considered together before the enterprise could responsibly move beyond planning.

6. Decisions the research changed

Reconsidering the initial sequence

The starting proposition favored blocks as a feasible entry point. Community feedback and wider research led me to consider establishing MRV earlier. The value of that change was to bring measurement questions into the planning process before claiming environmental outcomes. This was an evolution of the blueprint, not a completed MRV deployment.

Challenging the full-campus proposition

A dated review on 4 July 2026 challenged the then-current full-campus financing and timing assumptions, including dependence on prospective carbon income and insufficiently established demand. The review did not validate the original plan. It prompted a narrower, staged approach to the first commitment.

Defining a smaller first stage in Kayunga

The subsequent Kayunga review considered a blocks-focused first stage with evidence gates ahead of major equipment commitments. Questions about soil, land verification and demand belonged before expenditure that would be difficult to reverse. The public record here shows those conditions were designed; it does not assert that each was subsequently satisfied.

This later staging decision and the earlier attention to MRV answer different questions. One concerns how early measurement requirements should shape the concept; the other concerns what a feasible first operating stage could be. The record shows a plan being tested and revised over time.

Separating production from carbon revenue

The July 2026 methodology and scale reviews examined the assumptions connecting materials output to potential emission reductions. Product mass, stabilizer proportions, energy and transport affect a model. The resulting discipline was to treat carbon revenue as conditional on methodology, evidence and verification, rather than as cash already available to fund the project.

Requiring operational detail before layout

The 17 July pre-layout work identified information needed before a useful site arrangement could be developed: product matrix, mixture basis, production assumptions, surveys and flow between receipt, preparation, curing, quality review and dispatch. This converted an attractive spatial concept into a list of questions that could be answered and checked.

These examples are the strongest public evidence of my approach: research changed scope, sequence and conditions for proceeding. None of them depends on claiming that a plant was completed.

7. Resources, expenditure and outputs

Research required sustained investment

I report more than $26,000 in R&D expenditure over more than two years. The historical account identifies this as R&D expense; the present report is a founder account, not an audit. It does not raise the total beyond that supported threshold or assign an unsupported dollar value to the research itself.

The spending covered labor and salaries, rented tools, district-level work and coordination, meetings with local leaders, and customary tokens and hospitality associated with those engagements. The categories explain what sustained field research required. No category totals, percentages, unit costs or precise allocation to individual districts are supplied.

Supporting money movement and records

Tools used across the work included Sendwave, Wise, Western Union, MTN MoMo and Airtel Money for money movement, and QuickBooks, Wave and spreadsheet ledgers for financial records. Their use does not mean every transaction passed through every platform or that these systems formed one reconciled accounting environment. No private account records are included in the public package.

Outputs supported by this account and the retained materials

- A deployed community questionnaire and response-capture approach.
- A proxy-led engagement framework, facilitator materials and revision loop.
- A regenerative blueprint informed by research and community priorities.
- Dated reviews that challenged feasibility assumptions and revised the proposed sequence.
- Carbon-methodology screening and pre-layout requirements identifying further evidence needed.

The fieldwork and spending particulars are my first-person account. The retained written artifacts support examination of the framework, requirements and decision logic. A reader should not interpret the existence of a blueprint as proof that every proposed activity occurred.

What makes this work distinctive

Its value lies in the connection between local participation and project decisions: designing an inquiry, helping others deliver it, interpreting what came back, and being willing to reduce or reorder the plan. The research can be discussed through actual choices and artifacts without claiming exclusive access to an entire field or assigning a speculative valuation to the knowledge produced.

8. From the research to Pro Project Manager

Applying the same questions in a new tool

Pro Project Manager (PPM) is the name of a separate project-management tool. Empyrean Roots provides a practical case for explaining why its core questions matter: what are we trying to achieve, who is involved, what must happen next and what evidence justifies the decision?

PPM was not used to run the historical fieldwork. The connection below maps lessons from that work to the tool's current personal-workspace structure. It is an explanatory mapping, not a claim that the private research records have been imported into the product or released as a public case library.

Project: define the purpose and success criteria

In Empyrean Roots, research had to make an ambitious concept specific enough to test. PPM provides places for purpose, success criteria and a target date. An illustrative case criterion is whether a facilitator can use the material independently. That example makes success observable without inventing a respondent target.

People: make responsibilities and communication explicit

The proxy system depended on who prepared material, who adapted it, who delivered it and how feedback returned. PPM records responsibilities, involvement, needs and a communication plan. The lesson is that assigning a task without accounting for the person's context leaves part of the work undefined.

Delivery: show prerequisites before commitments

The research reviews identified evidence needed before a larger commitment. PPM's current workspace supports tasks, milestones, owners, dates or conditions, status and a prerequisite per item. A proposed teaching example is to make a planning decision depend on the required review being completed. This is a simplified illustration, not a representation of a full industrial approval system.

Value: preserve the reason a plan changed

The blueprint's revisions are useful because the rationale can be explained. PPM provides intended outcomes, measures, observations and decisions with reasons. The connection is direct: a changed plan should retain the finding that prompted the change, so later reviewers can understand it.

Present scope

The current tool is a local, browser-saved personal workspace with project backup export/import. Shared collaboration and a public production release are not claimed. PPM is a product name, separate from the PMP credential; this report implies no PMI endorsement or certification.

9. Evidence, authorship and references

How to read the evidence

This public case study combines my first-person account, retained project descriptions and dated research artifacts. District names, tools and spending categories were confirmed for this edition on 16 October 2026. Those confirmations support an account of my experience; they do not substitute for an independently audited dataset or expenditure statement.

The retained artifact base includes the Uganda community operations framework; field-research and R&D project accounts; the regenerative blueprint; July 2026 feasibility and Kayunga sequencing reviews; July 2026 carbon-methodology screening and pre-layout requirements; and later conceptual campus-layout review. The public report summarizes selected material without releasing the full commercial blueprint, participant records or private financial detail.

Authorship and review

Simon Mukooza prepared the research questions and response-capture approach, coordinated proxy delivery and synthesized findings into the blueprint. Empyrean Roots is co-founded with Kera, and the broader work includes her contributions and those of local facilitators. AI supported drafting, research organization and structured review; responsibility for the claims in this account remains with the author.

This publication presents an original project case, not a peer-reviewed scientific paper, independent assurance opinion, investment prospectus or engineering certification.

Public technical references

1. XLSForm documentation. Spreadsheet-based form authoring and structure. https://xlsform.org/en/

2. UNFCCC, AMS-III.Z, version 6. Methodology investigated during the research; inclusion does not establish eligibility. https://cdm.unfccc.int/methodologies/DB/VLZZ1DVT1QI3KHZKSM6QECOAKNSCXZ/

3. Verra, Verified Carbon Standard program. Program context only; no registration or verification is claimed. https://verra.org/programs/verified-carbon-standard/

4. Gold Standard for the Global Goals, standards and requirements. Research reference only; no certification is claimed. https://globalgoals.goldstandard.org/

These sources were opened during preparation on 16 October 2026. They explain the technical context and do not independently verify my historical use of the tools or the venture's results.

Discuss the work

Simon Mukooza | mukoozasimon@gmail.com

Discussion can cover questionnaire design, local facilitation, remote handoffs, iterative planning, challenged assumptions and the connection to the PPM tool. The strongest account of the work is the trail from a question, through evidence, to a decision.